

Andrew Xiao

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EDUCATION

Rutgers University – New Brunswick

Sep 2024 – May 2028

Bachelor of Arts in Computer Science, Minor in Economics

EXPERIENCE

Software Engineering Intern

Jun 2025 – Aug 2025

Mad Realities

Manhattan, NY

- Built a TypeScript RAG SMS chatbot integrating Community.com webhooks, Cloudflare KV storage, and Gemini, reducing API response times by **50%** and delivering AI-driven SMS functionality to **3,000+** users.
- Optimized a Palantir Foundry analytics dashboard by implementing asynchronous time-series rendering and lazy loading, reducing load time by **80%** (5s → 1s) across *MadMetrics*.
- Led development of *The Love Channel* microsite using React, TypeScript, and Cloudflare Workers, integrating Firebase phone number onboarding, reaching **1,000+** unique visits and **24+** submissions.

PROJECTS

RU My Valentine | *FastAPI, Pydantic, SQLAlchemy, PostgreSQL, AWS EC2*

Oct 2025 – Feb 2026

- Designed a weighted vector compatibility algorithm that separates categorical constraints (gender, orientation) as hard pre-filters before computing preference similarity scores across quiz responses.
- Implemented rate-limiting and JWT authentication with role-based access for security enforcement
- Launched to **1,400+** active Rutgers students on Valentine's Day, serving as the sole matchmaking platform for Rutgers-exclusive student sign-ups.

Dynamic Universal Badminton Rating (DUBR) | *React Native, Redis*

Mar 2026 – Jun 2026

- Built a cross-platform rating app (web + iOS/Android) backed by a shared PostgreSQL database, enabling consistent skill tracking across devices.
- Designed a Glicko-2 based rating engine extended with a custom score-differential factor using tanh normalization, tracking per-player uncertainty and volatility across matches.

TFTMonitor | *FastAPI, Pydantic, psutil, WebSocket*

Feb 2026 – Mar 2026

- Built a backend service that enforces per-user playtime limits by detecting active TFT sessions via **psutil** process monitoring and LCU WebSocket end-of-game events, with race condition handling for process edge cases.

LEADERSHIP

Club Manager

Oct 2022 – Jul 2024

New Jersey Badminton Club

Montville, NJ

- Implemented a membership system that increased revenue flow by 20% and boosted membership growth and retention by 15%.
- Organized and managed 2 large-scale tournaments with 100+ participants each, ensuring smooth court allocation and efficient time management.
- Led renovation efforts by upgrading all courts from wooden floors to rubber mats, improving play quality and boosting customer satisfaction by 40%.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS

Frameworks/Libraries: React, Next.js, FastAPI, SQLAlchemy, Pydantic, Tailwind CSS

Developer Tools: Git, Github, Node.js, psutil, WebSocket, Palantir Foundry, Cloudflare Workers/D1, Firebase, Supabase, AWS EC2, Docker

Activities: HackRU, Collegiate Badminton Team (Division 1 tournaments)